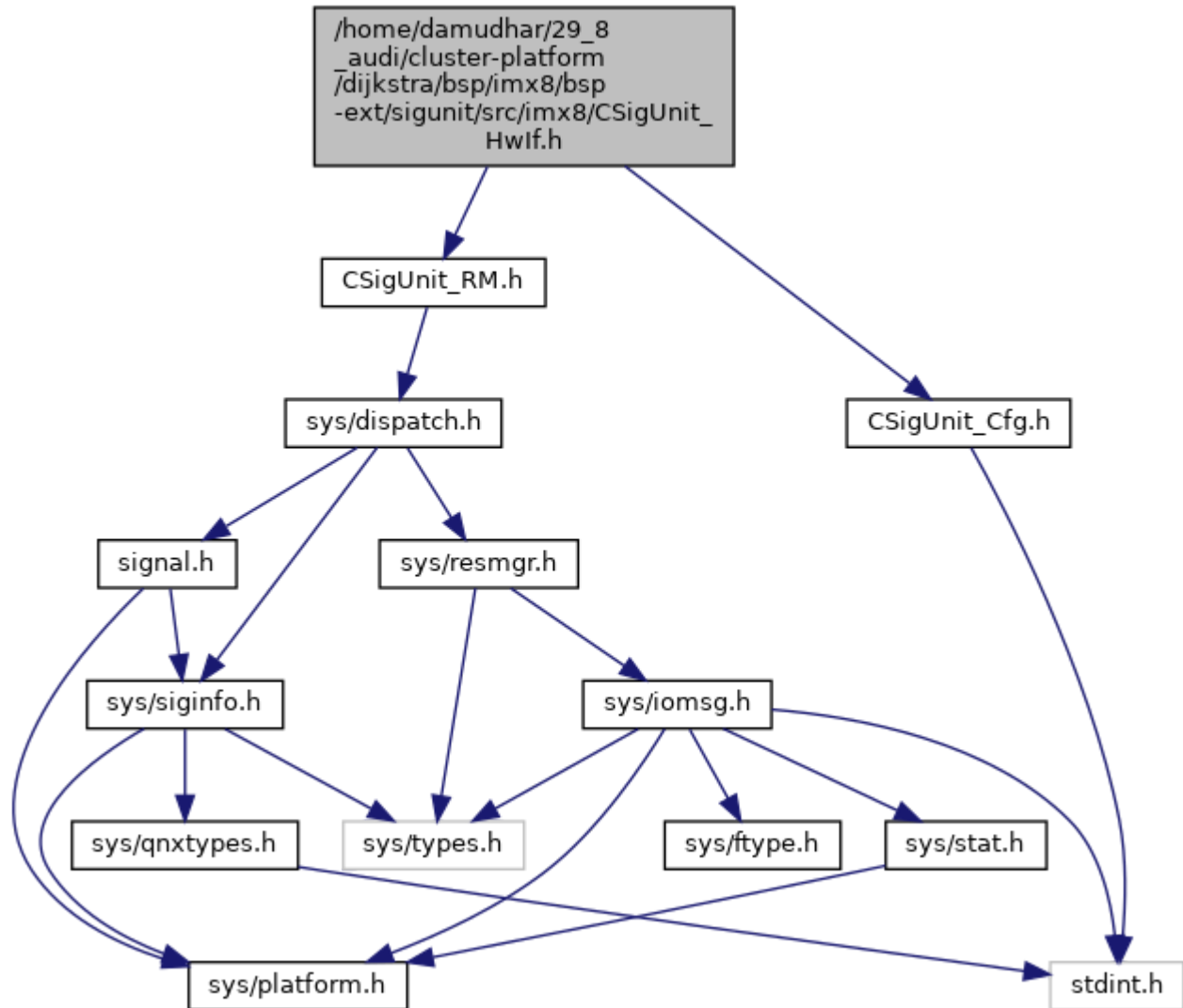


CSigUnit_Hwlf.h File Reference

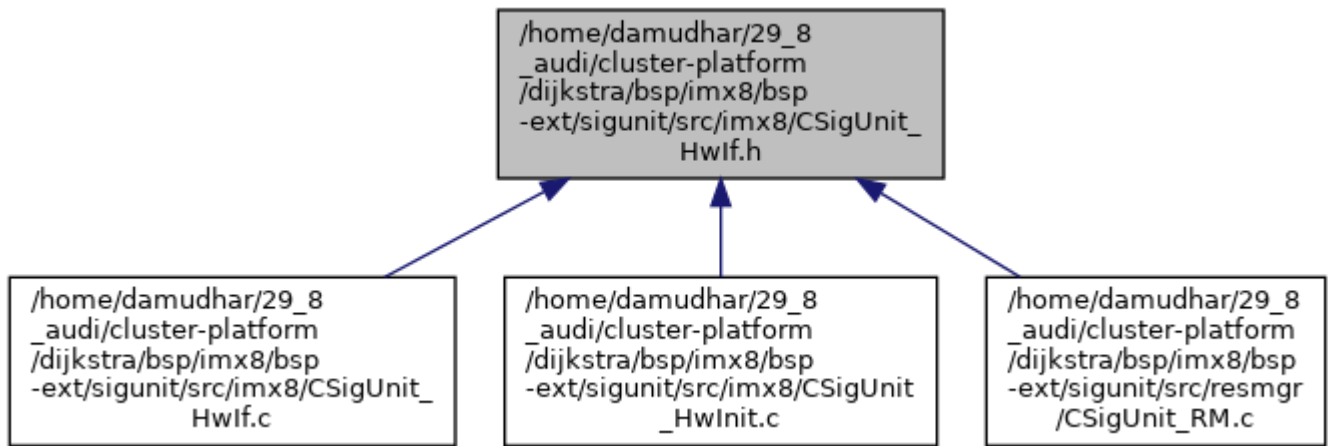
Interface to interact with the Signature Unit of iMX8. [More...](#)

```
#include "CSigUnit_RM.h"  
#include "CSigUnit_Cfg.h"
```

Include dependency graph for CSigUnit_Hwlf.h:



This graph shows which files directly or indirectly include this file:



[Go to the source code of this file.](#)

Classes

struct **ts_SSigUnitEvalWinReg_t**
For Windows setup macros. [More...](#)

Macros

```
#define SIG_UNIT_0 (0U)
#define SIG_UNIT_1 (1U)
#define SIG_UNIT_MAX (2U)
#define SIG_UNIT_EVAL_WIN_MAX (8U)
#define SIG_PANIC_COLOR_MASK ((uint32_t)0xFFFFF80)
#define SIG_IDLE_STATUS_MASK ((uint32_t)0x00100000)
#define SIG_ENABLE_EVWIN ((uint32_t)0x01)
#define SIG_ENABLE_CRC ((uint32_t)0x02)
#define SIG_ENABLE_ALPHAMASK ((uint32_t)0x100)
#define SIG_ENABLE_ALPHAINV ((uint32_t)0x200)
#define SIG_ENABLE_LOCALPANIC ((uint32_t)0x10000)
#define SIG_ENABLE_GLOBALPANIC ((uint32_t)0x20000)
#define SIG_WIN_CORDINATE_MASK ((uint16_t)0x3FFF)
```

Functions

int32_t **sigunit_initialize** (const **ts_sigunit_dev_t** *p_siginit_dev_SP, **ts_SSigUnitCfg_t** *p_SigUnitCfg_SP)
Function to initialize signature unit. [More...](#)

int32_t **sigunit_shutdown** (const **ts_sigunit_dev_t** *p_sigshutdowndev_SP)
Function to shutdown signature unit. [More...](#)

int32_t **sigunit_eval_win_setup** (const **ts_sigunit_dev_t** *p_sigsetupdev_SP, **ts_SSigUnitWinCfg_t** *p_SigUnitWinCfg_SP)

Function to setup and evaluation window. [More...](#)

int32_t **sigunit_eval_win_start** (const **ts_sigunit_dev_t** *p_sigevalstartdev_SP, uint8_t eval_win_id_win_start_U8)

Function to start and evaluation window. [More...](#)

int32_t **sigunit_eval_win_stop** (const **ts_sigunit_dev_t** *p_sigevalstopdev_SP, uint8_t eval_win_id_win_stop_U8)

Function to stop evaluation window by disabling continous mode. [More...](#)

int32_t **sigunit_eval_win_status** (const **ts_sigunit_dev_t** *p_sigevalstatusdev_SP, **ts_SSigUnitGetWinStatus_t** *p_SigUnitWinStatus_SP)

Function to get status of evaluation window. [More...](#)

bool **sigunit_hw_init** (**ts_sigunit_dev_t** *p_sighwinit_SP)

Function that is called to do IO Mapping with the SigUnit's Base Address. [More...](#)

bool **sigunit_hw_fini** (**ts_sigunit_dev_t** *p_sighwfini_SP)

Function that is called to do IO UnMapping with the banks having bits set more than 0. [More...](#)

Detailed Description

Interface to interact with the Signature Unit of iMX8.

This file provides functional defines for a Signature Unit module

Component Name: Signature Unit Archive: Date: May 11 2020

Author

:

Date	Author	Description
2020-08-02	RBABU9	Create Initial Module.
2023-09-13	damudhar	Added Doxygen document

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Macro Definition Documentation

◆ SIG_ENABLE_ALPHAINV

```
#define SIG_ENABLE_ALPHAINV ((uint32_t)0x200)
```

Bitmask for AlphaInv

◆ SIG_ENABLE_ALPHAMASK

```
#define SIG_ENABLE_ALPHAMASK ((uint32_t)0x100)
```

Bitmask for Alphamask

◆ SIG_ENABLE_CRC

```
#define SIG_ENABLE_CRC ((uint32_t)0x02)
```

Bitmask for EnCRC

◆ SIG_ENABLE_EVWIN

```
#define SIG_ENABLE_EVWIN ((uint32_t)0x01)
```

Bitmask for EnEvalWin)

◆ SIG_ENABLE_GLOBALPANIC

```
#define SIG_ENABLE_GLOBALPANIC ((uint32_t)0x20000)
```

Bitmask for EnGlobalPanic

◆ SIG_ENABLE_LOCALPANIC

```
#define SIG_ENABLE_LOCALPANIC ((uint32_t)0x10000)
```

Bitmask for EnLocalPanic

◆ SIG_IDLE_STATUS_MASK

```
#define SIG_IDLE_STATUS_MASK ((uint32_t)0x00100000)
```

◆ SIG_PANIC_COLOR_MASK

```
#define SIG_PANIC_COLOR_MASK ((uint32_t)0xFFFFF80)
```

Panic Colour Register is 25bits from MSB

◆ SIG_UNIT_0

```
#define SIG_UNIT_0 (0U)
```

◆ SIG_UNIT_1

```
#define SIG_UNIT_1 (1U)
```

◆ SIG_UNIT_EVAL_WIN_MAX

```
#define SIG_UNIT_EVAL_WIN_MAX (8U)
```

◆ SIG_UNIT_MAX

```
#define SIG_UNIT_MAX (2U)
```

◆ SIG_WIN_CORDINATE_MASK

```
#define SIG_WIN_CORDINATE_MASK ((uint16_t)0x3FFF)
```

14LSBs

Function Documentation

◆ sigunit_eval_win_setup()

```
int32_t sigunit_eval_win_setup ( const ts_sigunit_dev_t * pDev,  
                                ts_SSigUnitWinCfg_t * p_SigUnitWinCfg_SP  
                                )
```

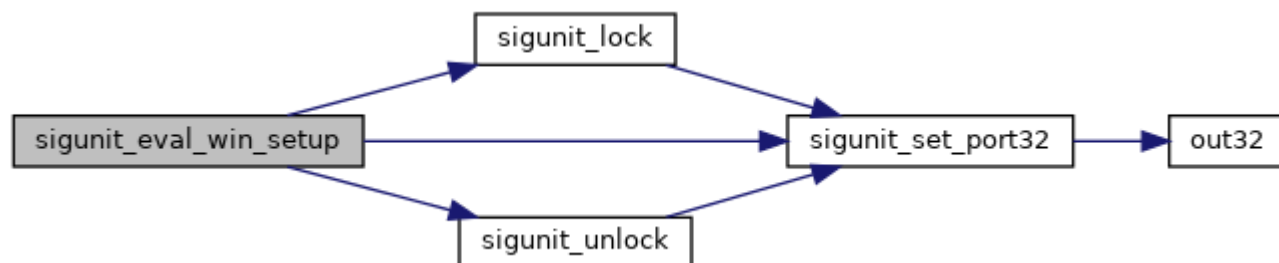
Function to setup and evaluation window.

Returns

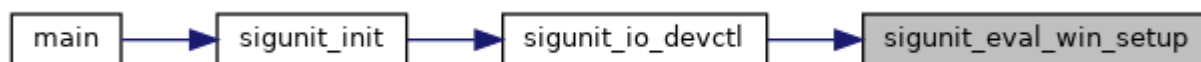
l_winsetup_status_S32

requirement ID 744947,744958

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_eval_win_start()

```
int32_t sigunit_eval_win_start ( const ts_sigunit_dev_t * pDev,  
                                uint8_t eval_win_id_U8  
                                )
```

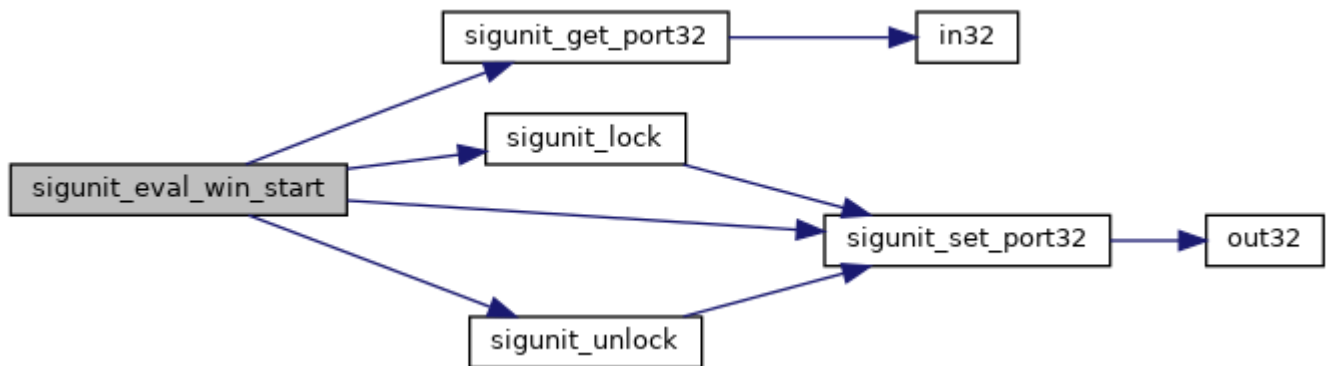
Function to start and evaluation window.

Returns

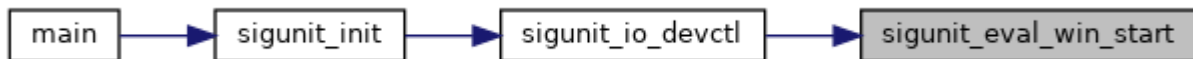
I_winstart_status_S32

requirement ID 744947,744959

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_eval_win_status()`

```
int32_t sigunit_eval_win_status ( const ts_sigunit_dev_t *    pDev,
                                ts_SSigUnitGetWinStatus_t * p_SigUnitWinStatus_SP
                                )
```

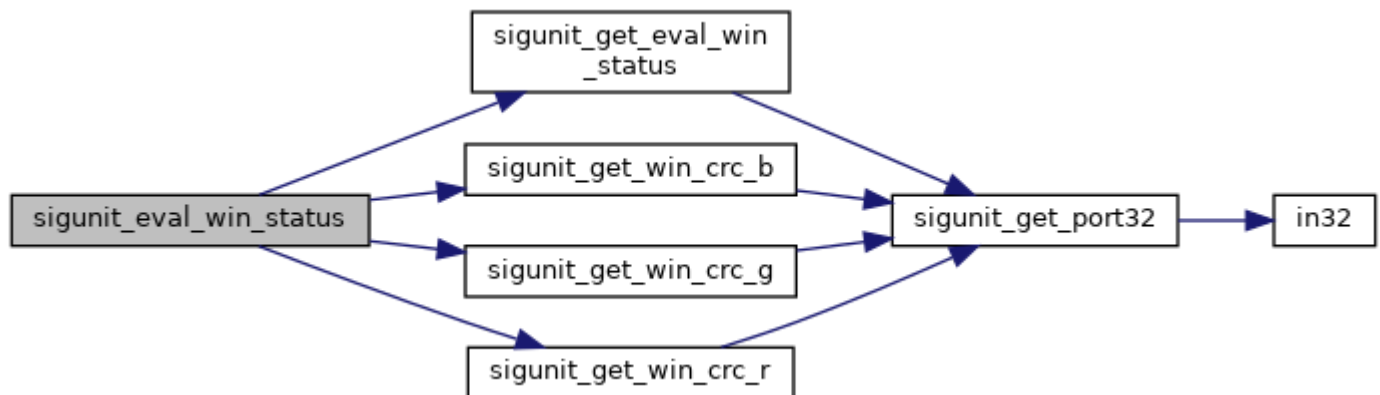
Function to get status of evaluation window.

Returns

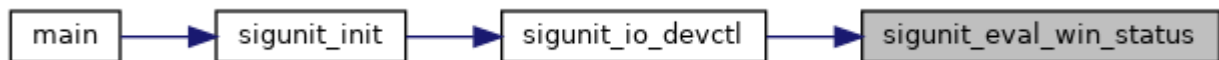
l_winstatus_status_S32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_eval_win_stop()`

```
int32_t sigunit_eval_win_stop ( const ts_sigunit_dev_t * pDev,
                               uint8_t                eval_win_id_U8
                               )
```

Function to stop evaluation window by disabling continous mode.

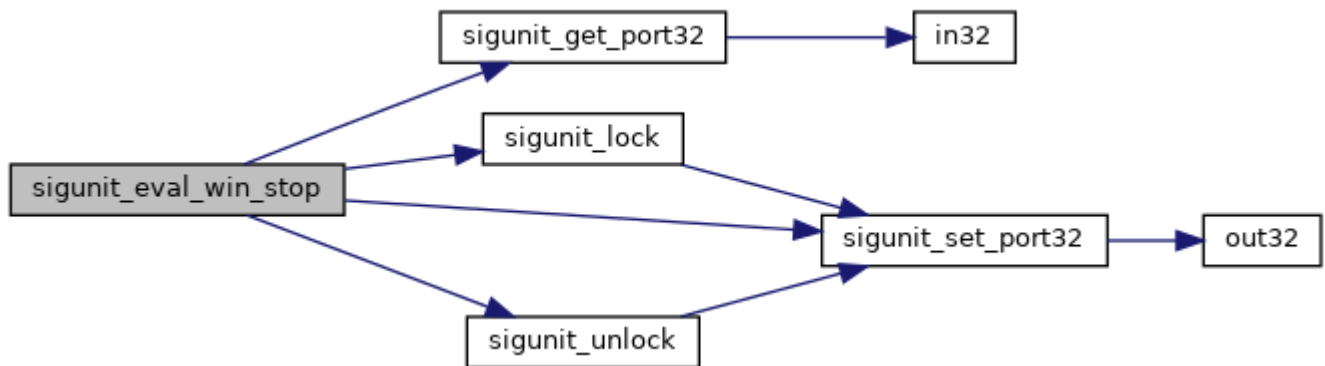
Returns

l_winstop_status_S32

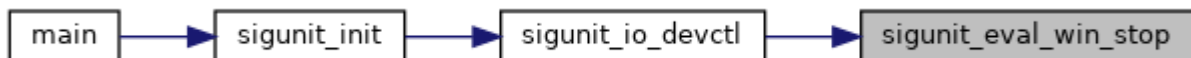
requirement ID 744947,744960

```
sigunit_set_port32( pDev->regbase + (uintptr_t) DISENG_SIG_CONTINUOUSMODE, (uint32_t) 0u);
```

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_hw_fini()`

```
bool sigunit_hw_fini ( ts_sigunit_dev_t * p_sighwfini_SP )
```

Function that is called to do IO UnMapping with the banks having bits set more than 0.

Function Name: sig_hw_fini

Parameters

ts_sigunit_dev_t *sighwfini_p

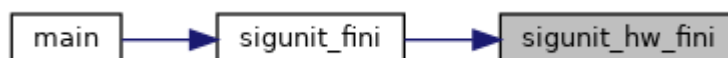
Returns

bool

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_hw_init()

```
bool sigunit_hw_init ( ts_sigunit_dev_t * p_sighwinit_SP )
```

Function that is called to do IO Mapping with the SigUnit's Base Address.

Function Name: sig_hw_init

Parameters

const **ts_sigunit_dev_t** *sighwinit_p

Returns

bool

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_initialize()

```
int32_t sigunit_initialize ( const ts_sigunit_dev_t * pDev,
                           ts_SSigUnitCfg_t *      p_SigUnitCfg_SP
                           )
```

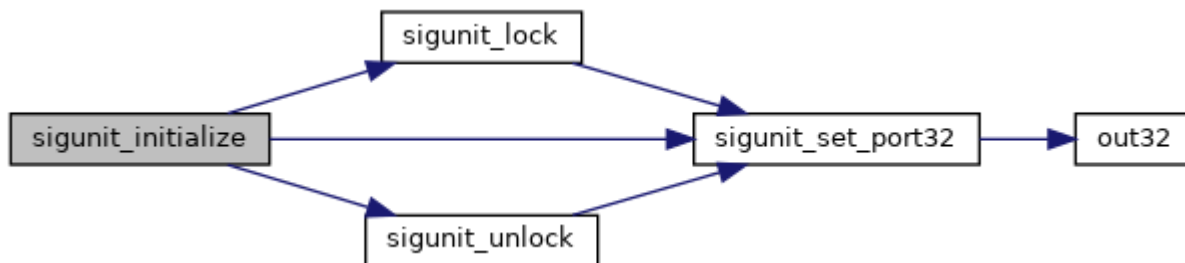
Function to initialize signature unit.

Returns

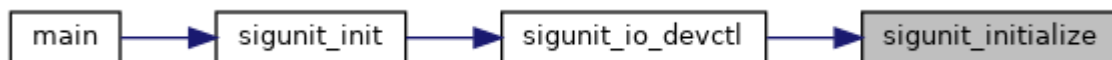
I_sigInit_status_S32

requirement ID 744947,744957

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_shutdown()

```
int32_t sigunit_shutdown ( const ts_sigunit_dev_t * p_sigshutdowndev_SP )
```

Function to shutdown signature unit.

Returns

I_sigshutdown_status_S32

requirement ID 744947,744953